

Dongfang Zhao, Ph.D.

Education

University of Washington (UW), Seattle	Computer Science, PostDoc	2017
Illinois Institute of Technology, Chicago	Computer Science, PhD	2015
Emory University, Atlanta, Georgia	Computer Science, MS	2008
Katholieke Universiteit Leuven, Belgium	Statistics, Diploma	2005

Professional Experience

2026–now	Open-source Contributor, Meta/Facebook Faiss (CUDA kernel)
2025–now	Open-source Contributor, Microsoft DiskANN (C++ main branch)
2023–now	Assistant Professor, UW Tacoma School of Engineering & Technology
2023–now	Affiliate Faculty, UW eScience Institute
2018–2023	Visiting Professor, University of California, Davis
2018–2023	Assistant Professor, University of Nevada, Reno (UNR)
2015–2015	Research Intern, IBM Almaden Research Center, San Jose, CA
2014–2014	Research Intern, Argonne National Laboratory, Chicago, IL
2013–2013	Research Intern, Pacific Northwest National Laboratory, Seattle, WA
2009–2011	Software Developer (full-time), Epic Systems Corporation, Madison, WI

Selected Awards

2022	Federal Government Research Grant, UNR PI, NSF 2112345, \$255,916 “A DLT Machine Learning Platform for Blockchain Warehousing”
2020	Federal Government Research Grant, UNR PI, DOE SC0020455, \$200,000 “HPChain: Energy-Efficient Scalable Blockchains for HPC Systems”
2019	Best Paper Award, International Conference on Cloud Computing “BlockLite: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud”
2018	Best Student Paper Award, IEEE International Conference on Cloud Computing “Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory”
2016	Postdoctoral Fellowship, Sloan Foundation, \$155,000
2007	Graduate Fellowship, Oak Ridge Institute for Science and Education, \$85,000

Recent Projects

[EGA]	Euclidean geodesic alignment for self-limiting adapters of frozen encoders perserving out-of-distribution geometry in vector search: https://github.com/hpdc/EGA Tags: Representation learning, Parameter-efficient fine-tuning (PEFT), SigLIP, DINOv2
[ElasticIVF]	Elastic in-memory vector indexing and search for streaming data analysis on heterogeneous distributed platforms: https://github.com/hpdc/ElasticIVF Tags: Distributed GPU computing, Message passing interface (MPI), Faiss, CUDA kernel
[AdaDisk]	Decentralized agentic systems for adaptive ingestion-query scheduling of disk-resident vector indices in LLM serving: https://github.com/hpdc/AdaDisk Tags: Large language models, AI agents, Distributed systems, DiskANN, C++, Python

Selected Publications (Dongfang Zhao was/is the *student's academic advisor)

- [1] *Han Zhang, **Dongfang Zhao**. RAE: A Neural Network Dimensionality Reduction Method for Nearest Neighbors Preservation in Vector Search. The 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2026.
- [2] **Dongfang Zhao**. SIVF: GPU-Resident IVF Index for Streaming Vector Analytics. The 35th ACM International Symposium on High-Performance Parallel and Distributed Computing (HPDC), 2026.
- [3] *Jiuzhou Fu, **Dongfang Zhao**. QPAD: Quantile-Preserving Approximate Dimension Reduction for Nearest Neighbors Preservation in High-Dimensional Vector Search. The 42nd IEEE International Conference on Data Engineering (ICDE), 2026.
- [4] *Jiayao Wang, ..., **Dongfang Zhao**. BadRSSD: Backdoor Attacks on Regularized Self-Supervised Diffusion Models. The IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Findings, 2026.
- [5] *Baigang Chen, **Dongfang Zhao**. Reliable Non-Leveled Homomorphic Encryption for Web Services. The ACM Web Conference (WWW), 2026.
- [6] *Chengyu Gong, *Gefei Shen, Luanzheng Guo, Nathan Tallent, **Dongfang Zhao**. Order-Preserving Dimension Reduction for Multimodal Semantic Embedding. The 40th Annual AAAI Conference on Artificial Intelligence (AAAI), 2026.
- [7] *Jiuzhou Fu, Luanzheng Guo, Nathan Tallent, **Dongfang Zhao**. ProHD: Projection-Based Hausdorff Distance Approximation. IEEE International Conference on Data Mining (ICDM), 2025.
- [8] *Jiayao Wang, ..., **Dongfang Zhao**. IPBA: Imperceptible Perturbation Backdoor Attack in Federated Self-Supervised Learning. European Conference on Artificial Intelligence (ECAI), 2025.
- [9] *Gefei Shen, **Dongfang Zhao**. Towards Nonparametric Topological Layers in Neural Networks. The 28th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD), 2024.
- [10] *Muhammad Jahanzeb Khan, ..., **Dongfang Zhao**. Privacy-Preserving Artificial Intelligence on Edge Devices: A Homomorphic Encryption Approach. IEEE International Conference on Web Services (ICWS) 2024: 395-405.
- [11] *Olamide Timothy Tawose, Jun Dai, Lei Yang, **Dongfang Zhao**. Toward Efficient Homomorphic Encryption for Outsourced Databases through Parallel Caching. Proceedings of the ACM on Management of Data (SIGMOD), 2023.
- [12] *Abdullah Al-Mamun, **Dongfang Zhao**. ECHAIN: Securing Electronic Device Provenance through Privacy-Preserving Consortium Blockchains. The 53rd Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2023.
- [13] *Xinying Wang, ..., **Dongfang Zhao**. Memory Scaling of Cloud-based Big Data Systems: A Hybrid Approach. IEEE Transactions on Big Data (TBData). 8(5): 1259-1272 (2022).
- [14] *Abdullah Al-Mamun, Haoting Shen, **Dongfang Zhao**. DEAN: A Lightweight and Resource-efficient Blockchain Protocol for Reliable Edge Computing. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022.
- [15] *Olamide Timothy Tawose, *Bin Li, Lei Yang, Feng Yan, **Dongfang Zhao**. Topological Modeling and Parallelization of Multidimensional Data on Microelectrode Arrays. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022.
- [16] *Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. SciChain: Blockchain-enabled Lightweight and Efficient Data Provenance for Reproducible Scientific Computing. 37th IEEE International Conference on Data Engineering (ICDE), 2021.

- [17] *Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. BAASH: lightweight, efficient, and reliable blockchain-as-a-service for HPC systems. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2021.
- [18] *Xinying Wang, *Olamide Timothy Tawose, Feng Yan, **Dongfang Zhao**. HDK: Toward High-Performance Deep-Learning-Based Kirchhoff Analysis. The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI), 2020.
- [19] **Dongfang Zhao**, Mohamed Mohamed, Heiko Ludwig. Locality-Aware Scheduling for Containers in Cloud Computing. IEEE Transactions on Cloud Computing (TCC). 8(2): 635-646 (2020).
- [20] Heyang Qin, Syed Zawad, Yanqi Zhou, Lei Yang, **Dongfang Zhao**, Feng Yan. Swift machine learning model serving scheduling: a region based reinforcement learning approach. The International Conf. for High Performance Computing, Networking, Storage and Analysis (SC), 2019.
- [21] Parmita Mehta, Sven Dorkenwald, **Dongfang Zhao**, Tomer Kaftan, Alvin Cheung, Magdalena Balazinska, Ariel Rokem, Andrew J. Connolly, Jacob VanderPlas, Yusra AlSayyad. Comparative Evaluation of Big-Data Systems on Scientific Image Analytics Workloads. Proceedings of Very Large Data Bases (VLDB), 2017.
- [22] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Zhihan Lu, Xiaohua Xu. Toward Efficient and Flexible Metadata Indexing of Big Data Systems. IEEE Transactions on Big Data (TBData). 3(1): 107-117 (2017).
- [23] **Dongfang Zhao**, Kan Qiao, Jian Yin, Ioan Raicu. Dynamic Virtual Chunks: On Supporting Efficient Accesses to Compressed Scientific Data. IEEE Transactions on Services Computing (TSC). 9(1): 96-109 (2016).
- [24] **Dongfang Zhao**, Ning Liu, Dries Kimpe, Robert B. Ross, Xian-He Sun, Ioan Raicu. Towards Exploring Data-Intensive Scientific Applications at Extreme Scales through Systems and Simulations. IEEE Transactions on Parallel and Distributed Systems (TPDS). 27(6): 1824-1837 (2016).
- [25] Tonglin Li, Xiaobing Zhou, Kevin Brandstatter, **Dongfang Zhao**, Ke Wang, Anupam Rajendran, Zhao Zhang, Ioan Raicu. ZHT: A Light-Weight Reliable Persistent Dynamic Scalable Zero-Hop Distributed Hash Table. IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2013.
- [26] **Dongfang Zhao**, Li Yang. Incremental Isometric Embedding of High-Dimensional Data Using Connected Neighborhood Graphs. IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI). 31(1): 86-98 (2009).